Paper Title

- A Generalized Dual Simplex Method for Solving Deconvolution Problem.
-

Fig 1. By variance share redistribution, Sinkhorn scaling can cause poor subspace estimation and limit the deconvolution performance of the original DualSimplex method.

A) DualSimplex methods find solution vertices in the projected space of Sinkhorn scaled matrix.

B) Different noise model changes simplex geometry in the projected space differently.

C) DualSimplex method has different deconvolution accuracy under different noise models.

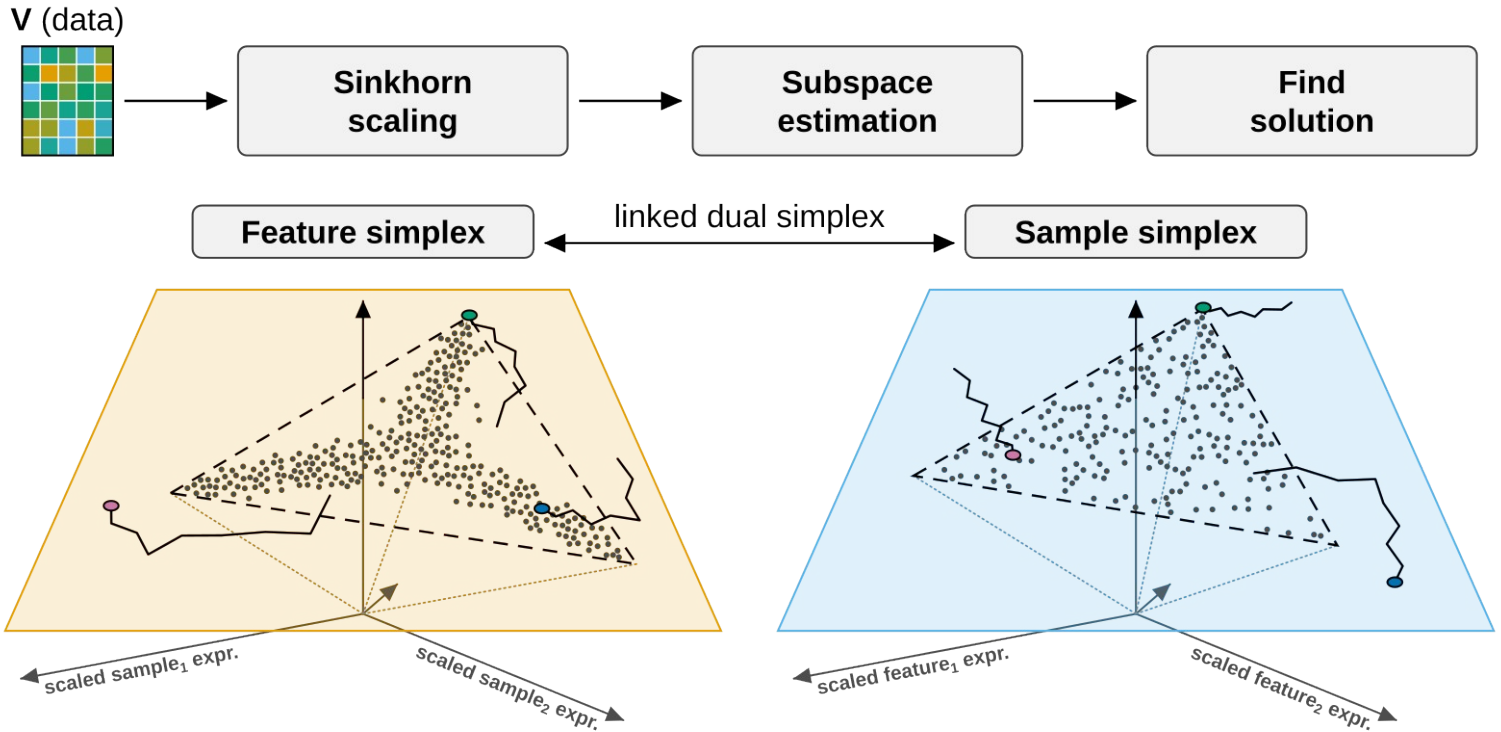
D) Sinkhorn scaling increases the variance share of low-expression features.

E) Variance share redistribution changes the signal-to-noise ratio (SNR).

F) Decreased SNR is associated with poor spectral separation by singular value decomposition (SVD).

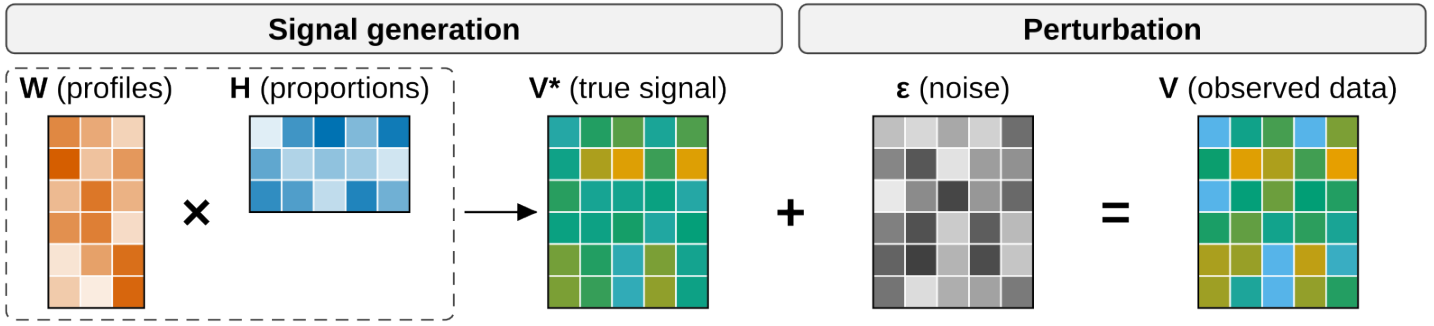
G) Poor SVD subspace prevents simplex vertices recovery.

Fig 1A. DualSimplex methods find solution vertices in the projected space of Sinkhorn scaled matrix.



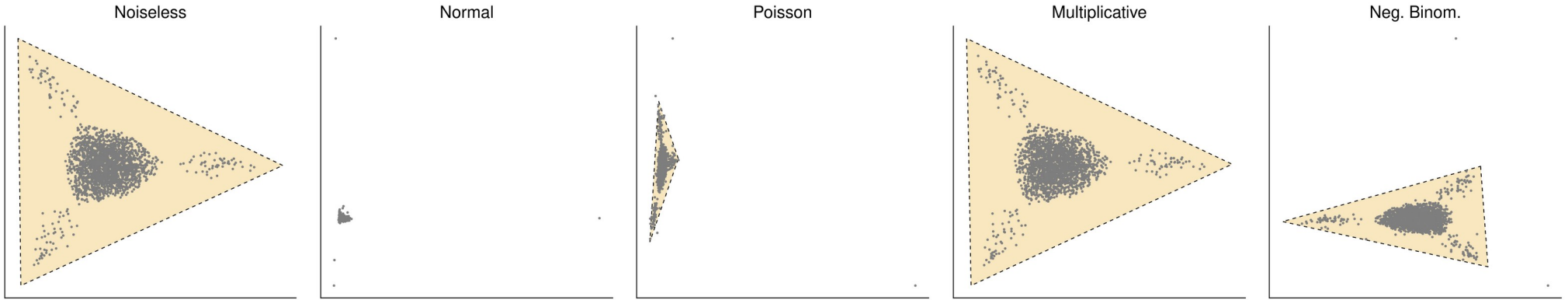
(A) DualSimplex solves deconvolution in low-dimensional subspaces of Sinkhorn-scaled data. It uses the linked geometry of feature and sample simplices induced by Sinkhorn scaling. Dashed triangles show the theoretical simplices of the true signal. Colored dots represent inferred vertices, and black lines trace their optimization trajectories.

Fig 1B. Different noise model changes simplex geometry in the projected space differently.



Noise	Variance type
Normal	constant‡
Poisson	linear (L)
Mul	quadratic (Q)
NB	L + Q

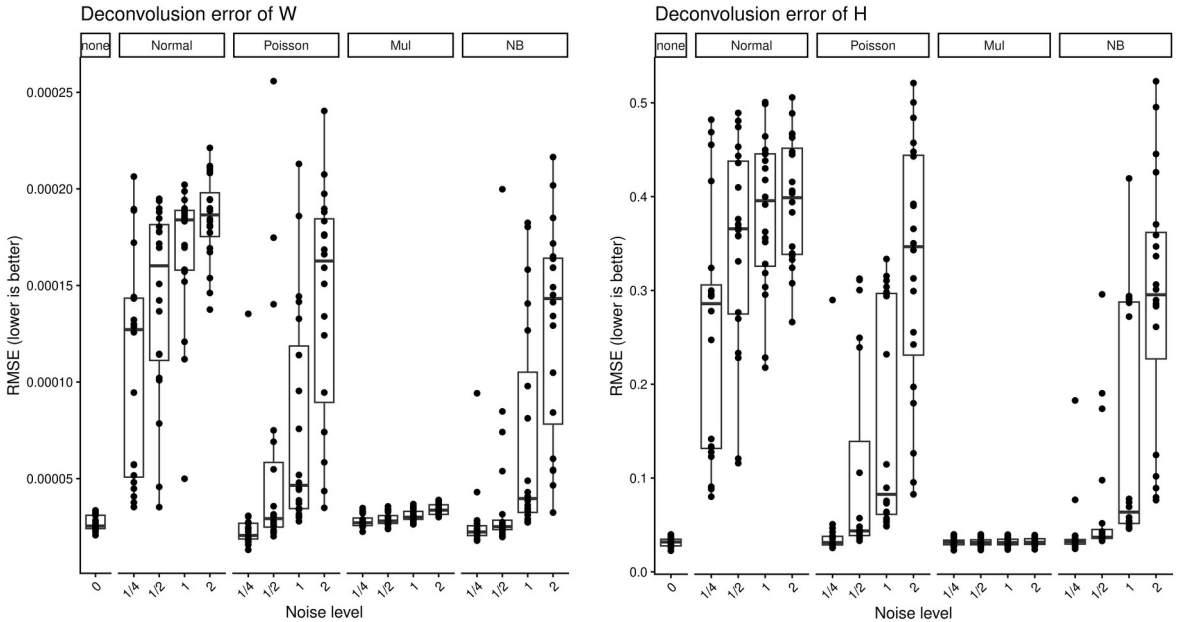
Example feature simplices from the same signal



(B) Perturbation experiment with four noise models differing in how their variance depends on the true signal. Example feature simplices show the same signal perturbed at equal noise levels. Colored triangles indicate the true simplex. Mul: multiplicative log-normal; NB: negative binomial. ‡ Normal noise is clipped to keep observed data non-negative.

Fig 1C. DualSimplex method has different deconvolution accuracy under different noise models.

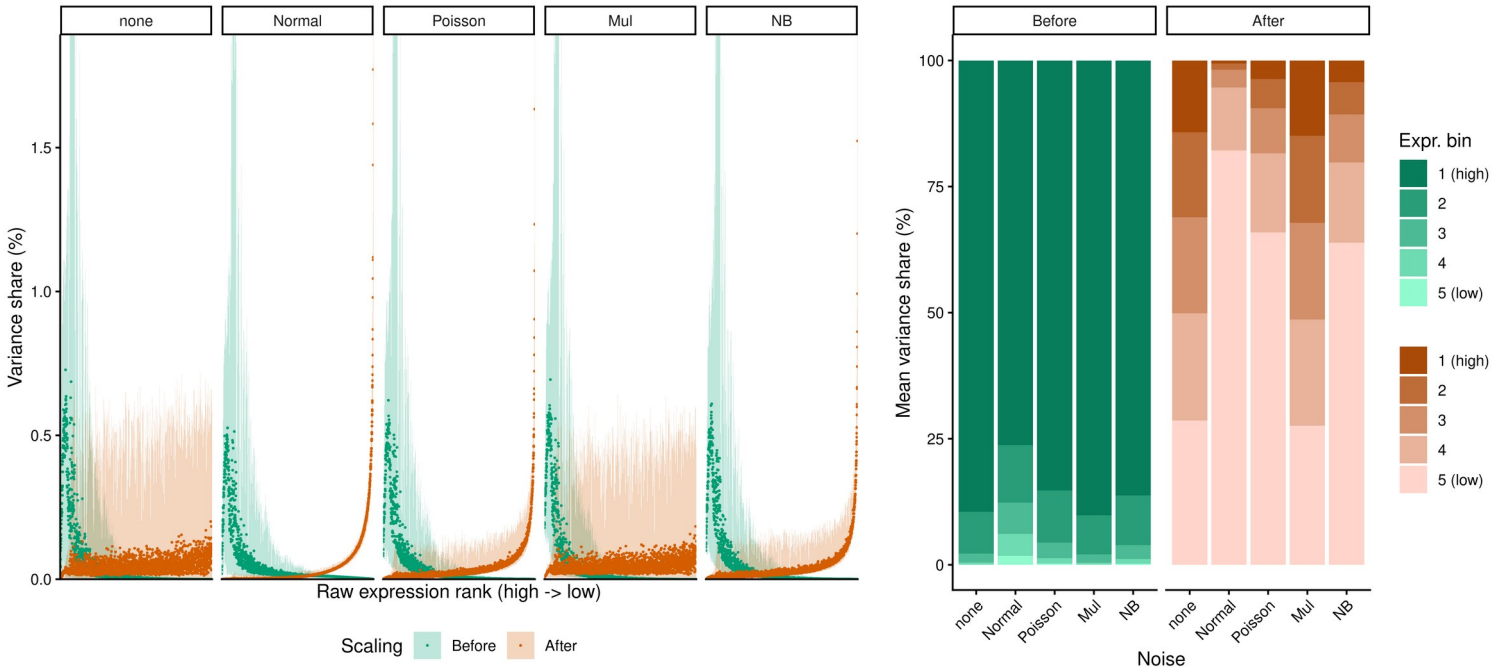
$$Noise\,level(L) = \frac{\sum_{i=1..m} Var[\epsilon_{i,.}|V^*_{i,.}]}{\sum_{i=1..m} Var[V^*_{i,.}]} = \frac{\text{total perturbation variance in features}}{\text{total signal variance in features}}$$



(C) Among tested noise models, DualSimplex is most robust to multiplicative log-normal noise, maintaining accuracy near the noiseless baseline even when noise variance is twice the signal variance. Metrics are calculated in the original expression space after reverse Sinkhorn procedure, with columns of W sum-to-one normalized.

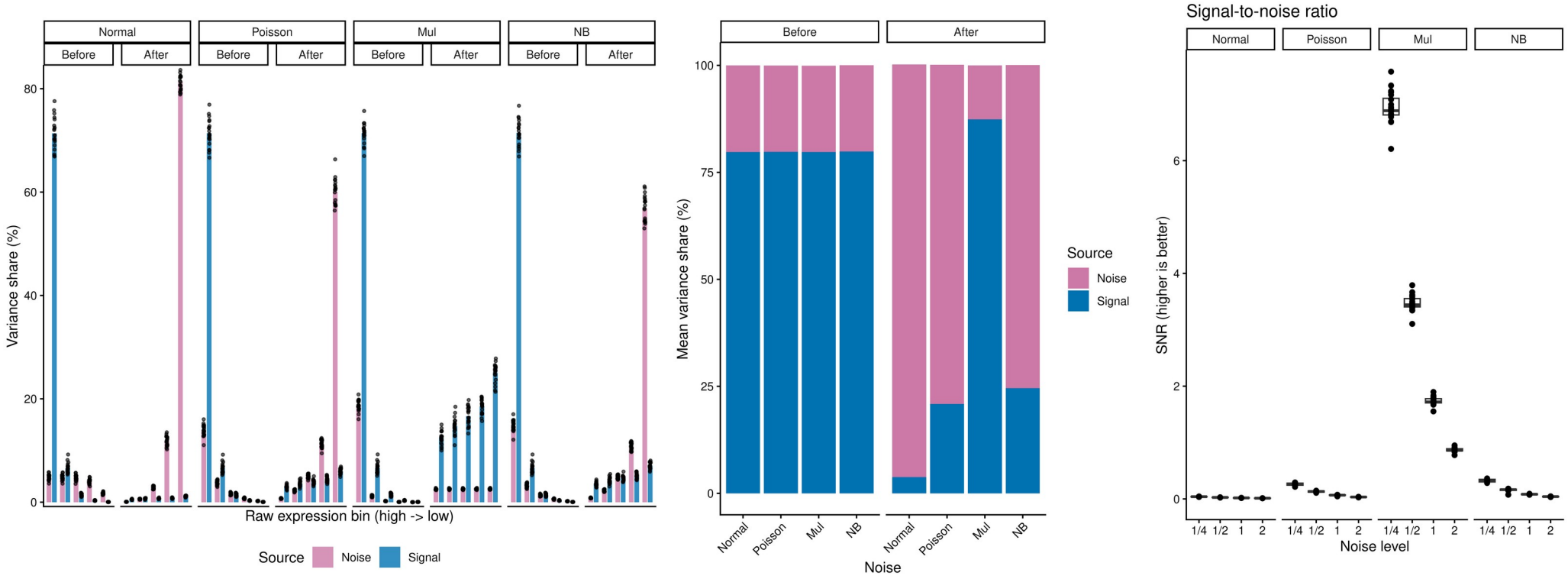
Fig 1D. Sinkhorn scaling increases the variance share of low-expression features.

$$\textit{Variance share of feature}_i: \frac{Var[V_{i,.}]}{\sum_{i=1..m} Var[V_{i,.}]} \text{ (raw) and } \frac{Var[(D_r V D_c)_{i,.}]}{\sum_{i=1..m} Var[(D_r V D_c)_{i,.}]} \text{ (scaled)}$$



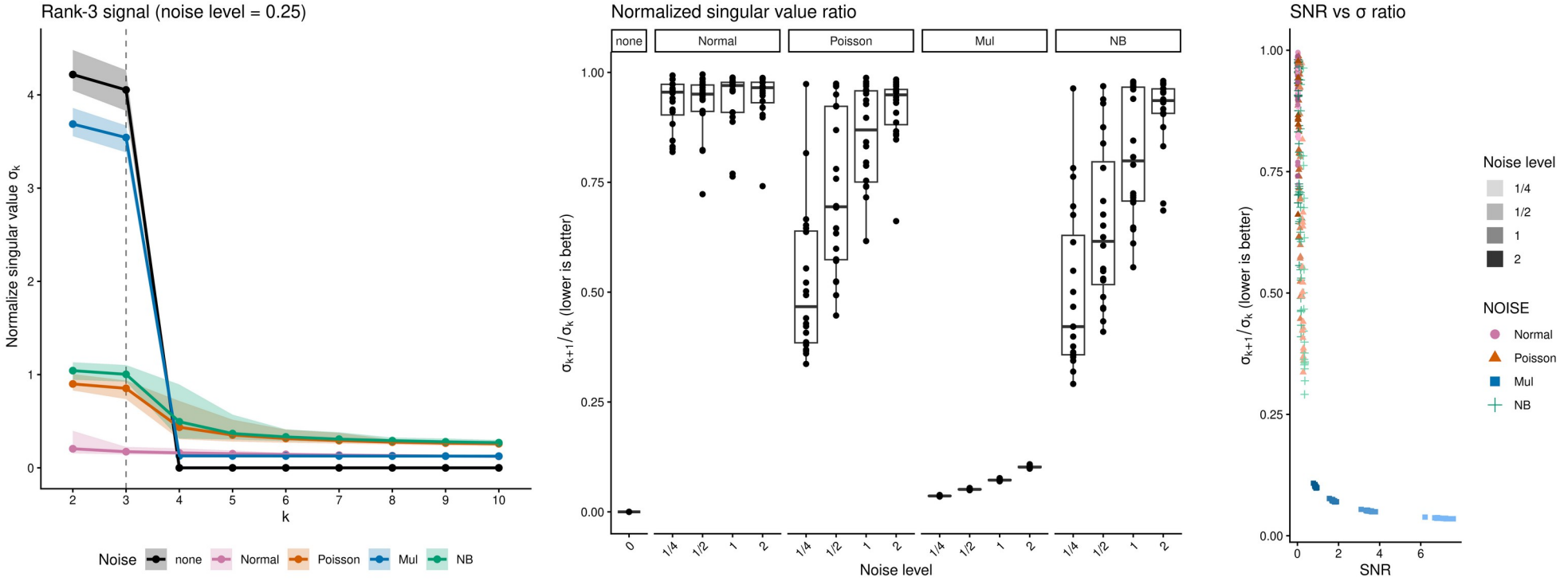
(D) Sinkhorn scaling increases the variance share of low-expression features. Points and shading show the mean and full range of variance shares at each raw expression rank. Bars summarize variance shares by expression bin, with similar numbers of features per bin. D_r and D_c are the Sinkhorn scaling matrices for V. Noise level: 0.25.

Fig 1E. Variance share redistribution changes the signal-to-noise ratio (SNR).



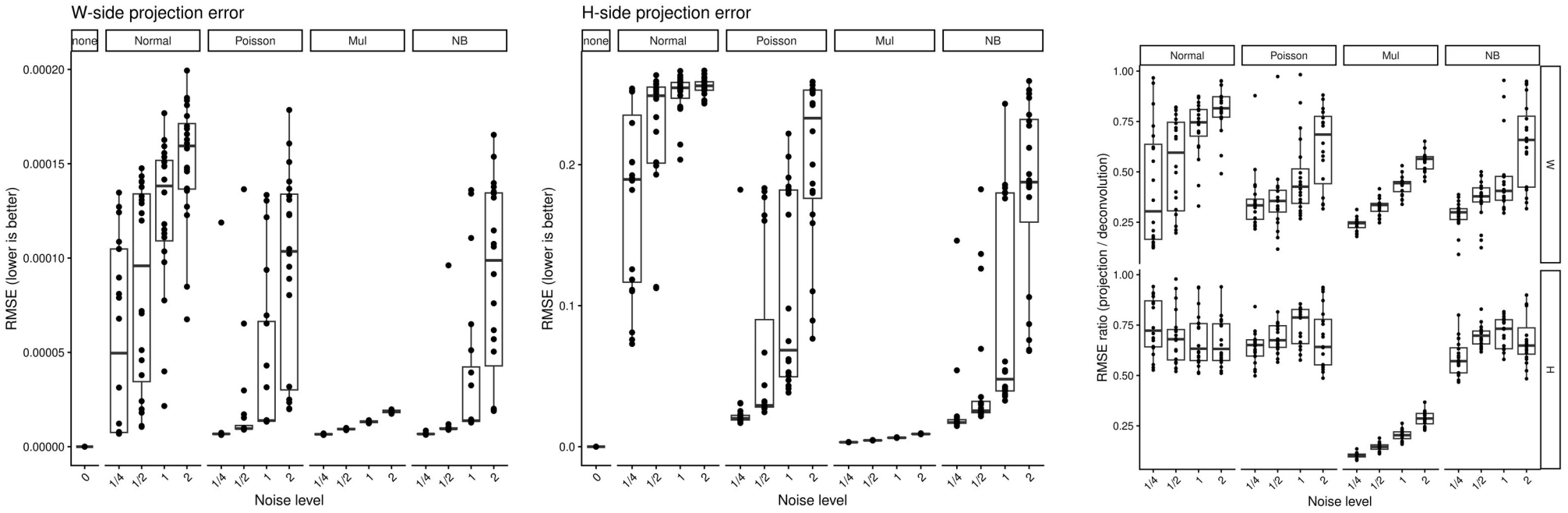
(E) Variance redistribution increases the noise share of total observed variance, except under multiplicative noise. Shares are calculated as in D, using variances of V* and ε before scaling and $D_r V^* D_c$ and $D_r \varepsilon D_c$ after scaling in the numerator. Covariance term is omitted from the visualization. Left and center: noise level 0.25. Right: signal-to-noise ratio across noise levels.

Fig 1F. Decreased SNR is associated with poor spectral separation by singular value decomposition.



(F) Except under multiplicative noise, perturbation drastically narrows the gap between σ_3 and σ_4. Lower post-scaling SNR is associated with weaker spectral separation. Left: mean normalized singular values across 20 simulations, with shading showing the full range, at noise level 0.25. Center: singular-value ratios across noise levels. Right: post-scaling SNR versus singular-value ratio.

Fig 1G. Poor SVD subspace limits simplex vertices recovery.



(G) Projection error increases with noise level when the true signal is projected onto SVD subspaces of noisy data. Projection-to-deconvolution RMSE ratios approach one in some cases, which indicating subspace estimation can limit vertex recovery. Metrics are calculated in the original expression space after reverse Sinkhorn procedure, with columns of W normalized to sum-to-one.